704 classes of ops.

Fixed-Point Operations

CLA	– load
ADD	add
ADM	add +(mod)
CLS	– load –
SUB	add +(mod)
SBM	add +(mod)
MPY, MPR	} mpy
RND	– special
DVH, DVP	} div with interrupt
LDQ	– load (full)
STQ	Store (full)
SLQ	Store (partial)
STO	Store (full
STP	Store (partial)
STD	Store (partial)
STA	Store (partial

Logical Operations

CLM	(STZ acc)
STZ	(storage clear)
CHS, SSP, SSM	} sign manipulators

CAL — logical CLA (P bit + sign)
SLW — logical STO

ANA, ANS } and — to Acc or to stg.
ORA, ORS } or
COM "not A"
ACL — for checksums

Shifting operations

ALS "off set"
ARS
LLS
LRS
LGL — take sign of Q
RQL — rotate Q

Floating Point:

FAD
UFA add unnormalized mod
FSB add + (mod)
UFS add + (mod)

FMP mpy
UFM Mpy + (mod)

FDH, FDP } interrupt

Logical connectives

A	0	0	1	1
B	0	1	0	1
result C	0	0	0	0
	0	0	0	1
	0	0	1	0
	0	0	1	1
	0	1	0	0
	0	1	0	1
	0	1	1	0
	0	1	1	1
	1	0	0	0
	1	0	0	1
	1	0	1	0
	1	0	1	1
	1	1	0	0
	1	1	0	1
	1	1	1	0
	1	1	1	1

Areas of improvement

1. Faster circuit components
2. Synchronous design
3. Control and I/O
4. Indicators + testing
5. Interrupt system
6. VFL generalized (prod fix)
7. Fl Point generalized (+ fix pt.)
8. Debugging
9. Checking.

(25 people)

Control operations:

NOP

HPR — halt & pr.

ETM — } Trapping mode - Type of interrupt
LTM — }

HTR — halt

TRA

TZE — } on & off — Special indicators on results
TNZ — }

TPL

TMI

TOV — } or "on & off"
TNO — }

TQP — Q plus.

TQO — Q overflow

TLQ — } "compare" opn. { Q vs A
CAS — } { A vs S

TSX — branch link

TXI — TIX'es

TXH

TXL

TIX

TNX

TTR — Trap transfer

PBT — p bit test

LBT — low bit test

DCT — Divide check
RTT — tape check — machine result failure

Index loading + storing

LXA
LXD
SXD
PAX
PDX
PXD

partial fields.

(no SXA + PXA)

Input-output

(RDS CPY
WRS)

BST - tape
WEF - tape
REW - tape
ETT - tape

LDA Drum

PSE } test switches
MSG } "on + off"

/

Property of